TEXORA · SESSIONAL NOTES · STUDY SHEET

Industrial Lock Stitch Machines and Stitch Samples

How the lock stitch (type 301) is formed, cycle by cycle, between needle thread and bobbin thread — and how to produce stitch samples, set the stitch length, measure stitches per inch, and read the common stitch defects from a faulty sample.

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1. Aim and place of the study

Aim: to study the formation of the lock stitch on the industrial single-needle lockstitch machine, produce stitch samples at set stitch lengths, measure their stitch density (SPI), and identify common stitch defects and their causes. Study 6 made the student familiar with the machine; this study explains what the machine is doing inside the fabric, where the eye cannot follow.

2. The lock stitch — what is formed

The **lock stitch (ISO stitch type 301)** is formed by two threads: the **needle thread** from the cone above, and the **bobbin thread** wound on the small bobbin below the needle plate. The needle thread is carried through the fabric as a loop, the loop is passed around the bobbin thread, and the two threads are pulled into the middle of the fabric plies where they interlace — "locked". A correct 301 seam looks the same on both faces: a straight line of thread segments, with the interlacing point hidden inside the material.

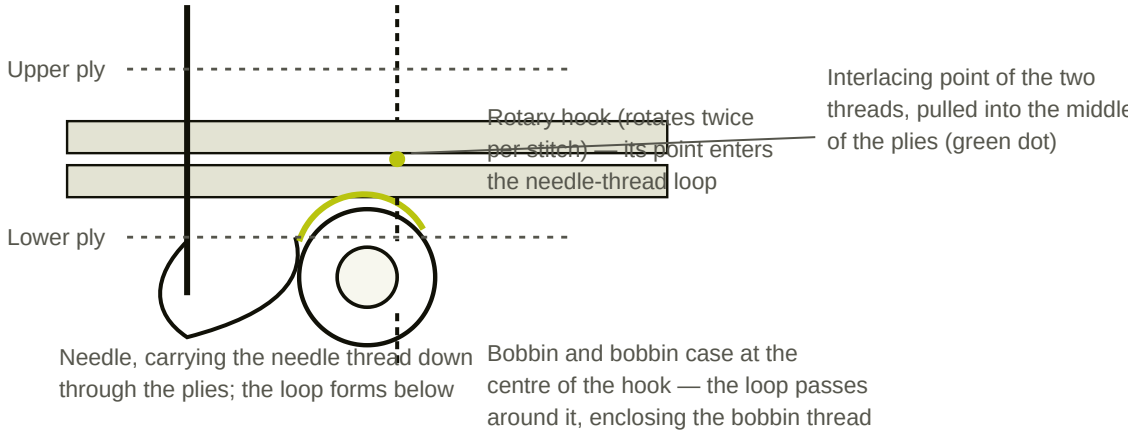


Figure 1 — The lock-stitch station in cross-section (schematic): needle and loop on the left, rotary hook and bobbin at centre, and the finished interlacing hidden inside the fabric plies.

3. Formation of the stitch, step by step

1. **Penetration.** The needle descends, carrying the needle thread through the fabric plies to its lowest point.
2. **Loop formation.** The needle rises about 2 mm from its lowest point. The thread, held by friction against the fabric, cannot rise with it, so a loop of thread bulges out on the scarf (back) side of the needle.
3. **Loop capture.** The point of the rotary hook — which rotates at twice the needle-bar speed, two revolutions per stitch — enters the loop precisely as it forms.
4. **Loop around the bobbin.** The hook carries the needle-thread loop around the bobbin case, so the loop encloses the bobbin thread; the needle meanwhile rises clear of the fabric.
5. **Stitch tightening.** The take-up lever rises sharply and pulls the loop up, drawing the bobbin thread into the fabric and the interlacing point into the middle of the plies.
6. **Feeding.** The feed dog moves the fabric by exactly one stitch length while the needle is out of the fabric, and the cycle repeats.

Every timing in the cycle is mechanical and fixed by the machine's setting. This is why a bent needle or a wrongly fitted needle (Study 6) shows up here as a *missed loop* — the hook point passes where the loop should be, and finds nothing.

4. Producing stitch samples and measuring SPI

1. Sew a straight seam of at least 15 cm on two plies of exercise fabric at the set stitch length, back-tacking at the ends.
2. Examine both faces: the sample passes when both faces show the same straight line of thread with no loops, and the interlacing is not visible on either face.
3. Measure the stitch density: lay a ruler along the seam, mark a length of exactly 1 inch (2.54 cm) — or count over a longer run and convert — and count the whole stitches inside the marks.

$$\text{SPI} = \text{Number of stitches counted} \div \text{Length measured (inches)}$$

Worked example. A sample shows 28 stitches over 7.0 cm of seam. Length in inches = 7.0 ÷ 2.54 = 2.76 in. SPI = 28 ÷ 2.76 ≈ **10 stitches per inch** (equivalently 28 ÷ 7 = 4 stitches per cm × 2.54 ≈ 10.2). Typical setting for shirt seaming is about 10–12 SPI: coarse enough to sew fast and unpick when needed, fine enough to hold the seam's strength and appearance. The specification — not habit — sets the SPI for each seam of a garment; too dense a stitching perforates and weakens lightweight fabric, too coarse gaps under strain.

5. Balancing the tensions — reading the sample

WHAT THE SAMPLE SHOWS	DIAGNOSIS
Loops of needle thread on the underside; top face correct	Needle-thread tension too loose (or bobbin tension too tight): the needle thread is not being pulled up into the fabric.
Bobbin thread showing as loops/dots on the top face	Needle-thread tension too tight (or bobbin tension too loose): the bobbin thread is dragged up to the surface.
Both faces straight lines, interlace hidden	Balanced — the target condition; adjust in small steps from here only if the fabric changes.
Seam puckers along the stitch line on both faces	Both tensions too high for the fabric, or presser-foot pressure/feeding problem — threads are contracting the fabric.

Adjust one tension at a time, in small steps, sewing a test length after each adjustment. The bobbin-case screw is set first and rarely touched; nearly all balancing is done on the numbered needle-thread tension dial (Study 6).

6. Common stitch defects

DEFECT	APPEARANCE	LIKELY CAUSES

Skipped stitch	A gap where one stitch is missing and the thread bridges straight across	Bent/blunt needle, needle fitted wrongly, loop not formed (wrong needle-thread combination), timing disturbed
Thread breakage	Seam stops with a frayed or snapped thread end	Excess tension, burr on needle eye or hook point, cheap/inconsistent thread, needle overheating on synthetics
Loose stitches	Loops standing on one face, seam can be pulled open by hand	Tension fault (see §5), thread not between the tension discs, bobbin case spring dirty or damaged
Puckering	Fabric gathers slightly along the seam	High tensions, dull needle displacing yarns, mismatched feeding of plies, too-dense SPI for the fabric
Needle damage marks	Holes, cuts or laddering in the fabric along the seam	Damaged needle point, needle too coarse for the fabric, sewing over pins
Uneven stitch length	SPI varies along one seam	Fabric dragged or held back by the operator, faulty feed dog, slipping under a worn presser foot

7. Precautions

- Change the needle before blaming the machine's timing: most "timing" complaints in a teaching laboratory are needle faults.
- Never sew over pins; remove each pin just before the needle reaches it.
- Keep fingers behind the needle line and use the guard; a lockstitch machine at production speed makes several thousand stitches a minute.
- Record every sample with its settings (needle size, thread, SPI, tension dial reading) — an unlabelled sample teaches nothing later.

8. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State the ISO stitch type number of the lock stitch and the two threads that form it.
2. Describe how the loop is formed below the fabric, naming the needle movement that creates it.
3. How many revolutions does the rotary hook make per stitch, and what does its point do?
4. A sample counts 36 stitches in 9.0 cm. Calculate its SPI.

5. A sample shows loops of needle thread underneath but a clean top face. Which tension is at fault and in which direction do you adjust the dial?
6. Give three causes of skipped stitches and the first check you would make.
7. Why is an excessively high SPI a defect on a lightweight shirting, not a sign of quality?

Answer key

1. Type 301. The needle thread (from the cone) and the bobbin thread (from the bobbin below the needle plate), interlaced in the middle of the plies.
2. After reaching its lowest point the needle rises about 2 mm; friction holds the thread against the fabric so it cannot rise with the needle, and a loop bulges out on the scarf side of the needle for the hook point to enter.
3. Two revolutions per stitch. On the working revolution its point enters the needle-thread loop and carries it around the bobbin case so the loop encloses the bobbin thread.
4. 9.0 cm = 3.54 in. SPI = 36 ÷ 3.54 ≈ 10.2 — about 10 SPI. (Or: 36 ÷ 9 = 4 per cm × 2.54 ≈ 10.2.)
5. The needle-thread tension is too loose (assuming the bobbin tension is at its set value): increase the needle-thread tension in small steps, testing after each step.
6. Any three of: bent or blunt needle; needle fitted incorrectly (groove/scarf orientation); thread too thick or thin for the needle so the loop forms badly; hook timing disturbed. First check: fit a new, correct needle — it cures the majority of cases.
7. Each needle penetration cuts or displaces yarns; a very dense stitching perforates a light fabric along the seam line like a tear-strip, so the seam fails by the fabric tearing, not the thread breaking — strength and appearance both suffer.
